Banoth Nikhil

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.64
narayana junior collage	Class XII (telangana state board of intermediate education)	2021	916.00%
paramita high school	Class X (board of secondary education)	2019	9.20%

PROJECTS

MCP Multi-Agent System | Python, LangGraph, MCP, FastAPI, Docker, GCP

- Built an end-to-end **multi-agent AI system** using LangGraph, coordinating specialized Coordinator, Research, Data, and Synthesis agents through a shared **Model Context Protocol (MCP) server**.
- Implemented reusable MCP tools for **knowledge-base search, sales-data retrieval, and report generation**, enabling multiple agents to access a standardized tool layer through a shared MCP client.
- Developed a **FastAPI REST endpoint** for executing agent workflows and returning execution traces, making agent routing and tool usage observable for debugging and demonstrations.
- Created a deterministic **pytest test suite** for MCP tools and routing logic, and integrated automated testing into a **GitHub Actions CI/CD pipeline** that builds Docker images and deploys the application to Google Cloud Run.
- Containerized the application with **Docker** and configured cloud deployment using **Google Artifact Registry and Cloud Run**, with deployment gated on successful automated tests.

Self-Correcting Code Assistant | Python, LangGraph, FastAPI, AWS Lambda, Docker, GitHub Actions

- Built a **self-correcting coding agent** using LangGraph that generates Python code from natural-language tasks, executes it, analyzes runtime tracebacks, and iteratively regenerates the code until successful or a maximum iteration limit is reached.
- Implemented an isolated **AWS Lambda execution sandbox** for running LLM-generated code separately from the orchestration API, preventing untrusted generated code from executing within the main application process.
- Developed a **FastAPI orchestration service** with configurable local and AWS Lambda sandbox backends and support for tracking each code-generation attempt, execution result, and traceback history.
- Created a deterministic **15-test Pytest suite** covering sandbox execution, Lambda integration logic, and agent retry behavior, and integrated automated testing into a **GitHub Actions CI/CD pipeline**.
- Containerized the orchestration service and deployed it using **AWS App Runner**, with Docker images pushed to Amazon ECR and deployment gated by successful automated tests.

SQL AI Assistant | Python, LangChain, Groq, FastAPI, Streamlit, Docker, GCP

- Built a **natural-language-to-SQL AI assistant** using a LangChain tool-calling agent that inspects database schema, generates SQL queries, executes them, and retries when query execution errors occur.
- Implemented **two-layer read-only database protection** using SQLite connection-level restrictions and SQL query validation to block write and destructive operations such as INSERT, UPDATE, DELETE, DROP, and ALTER.
- Developed a decoupled **FastAPI REST API** with a Streamlit frontend, separating the AI/query-processing backend from the user interface for easier deployment and maintenance.
- Created a **30-question evaluation framework** with ground-truth answers and edge-case tests covering ambiguous queries, missing data, and injection-style prompts to quantitatively evaluate agent performance.
- Containerized the API and UI using **Docker/Docker Compose** and deployed both services to **Google Cloud Run**, enabling independently scalable cloud-hosted access to the application.

SKILLS

Languages: Python, JavaScript, SQL

AI/LLM: LangGraph, LangChain, Multi-Agent Systems, LLM Tool Calling, Text-to-SQL, Self-Correcting Agents, Prompt Engineering

AI Protocols & APIs: Model Context Protocol (MCP), Gemini API, Groq API, REST APIs

Frameworks & Libraries: FastAPI, Streamlit, Pytest

Databases: SQLite

Cloud & DevOps: Docker, Docker Compose, GitHub Actions, Google Cloud Run, Google Artifact Registry, AWS Lambda, Amazon ECR, AWS App Runner

Developer Tools: Git, GitHub, Linux, Postman

POSITION OF RESPONSIBILITY

Teaching Assistant, Robotics, IITH

Assisted 100+ students by clarifying course concepts and evaluating assignments.